

2026 Programming Bootcamp, CSHL School of Biological Sciences

Date: 31 August 2026

Instructors:

- Justin Kinney, jkinney@cshl.edu
- Ivan Iossifov, iossifov@cshl.edu
- Hannah Meyer, hmeyer@cshl.edu

Teaching assistants:

- Leo Liu, zhliu@cshl.edu
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GitHub Repository:

Unix and Python Programming: https://github.com/jbkinney/26e_programming_bootcamp

R Programming I – IV: <https://github.com/meyer-lab-cshl/R-data-visualisation-course>

Important: Please complete **R Programming I** prior to the bootcamp – this part is installing software, which is outlined in detail [here](#); if any questions please reach out before the course

Day 1 (Mon 8/31, 1:30p-4:30p), afternoon: Plimpton Conference Room, Beckman Bldg.

1:30p: **Overview of Quantitative Biology (lecture, Justin)**

2:00p: **The Unix command line (tutorial, Ivan)**

Day 2 (Tue 9/1, 10:00a-1:00p), morning: James Library, James Bldg.

10:00a: **Introduction to Python and Jupyter Notebooks (tutorial, Justin)**

10:30a: **Python: data types (tutorial, Justin)**

11:45a: **Python: flow control (tutorial, Ivan)**

Day 2 (Tue 9/1, 2:00p-5:00p), afternoon: Plimpton Conference Room, Beckman Bldg.

2:00p: **Introduction to dataframes (lecture, Justin)**

2:15p: **Pandas I, TF analysis (tutorial, Ivan)**

3:45p: **NextGen sequencing and high-performance computing (lecture, Justin)**

4:00p: **Read mapping using Elzar (tutorial, Justin)**

Day 3 (Wed 9/2, 10:00a-1:00p), morning: Samet Conference Room, Koch Bldg.

10:00a: **R Programming II + III (tutorial, Hannah)**

Day 3 (Wed 9/2, 2:00p-4:30p), afternoon: Samet Conference Room, Koch Bldg.

2:00p: **Pandas II, Replication origin analysis (tutorial, Justin)**

3:30p: **Matplotlib (tutorial, Ivan)**

Day 4 (Thu 9/3, 10:00a-1:00p), morning: James Library, James Bldg.

10:00a: **R Programming IV (tutorial, Hannah)**